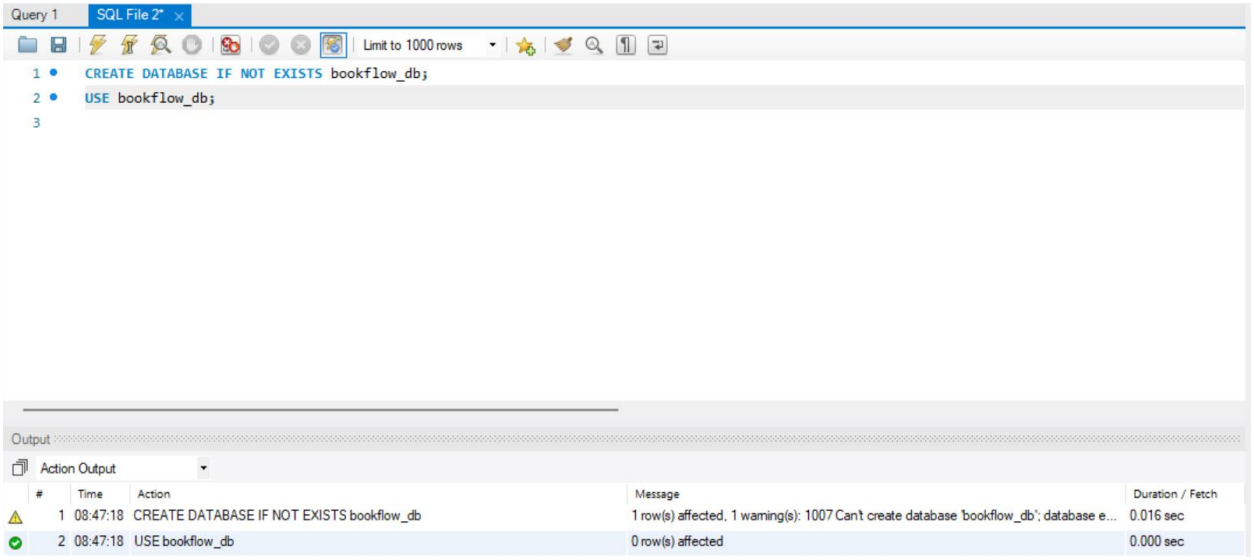
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

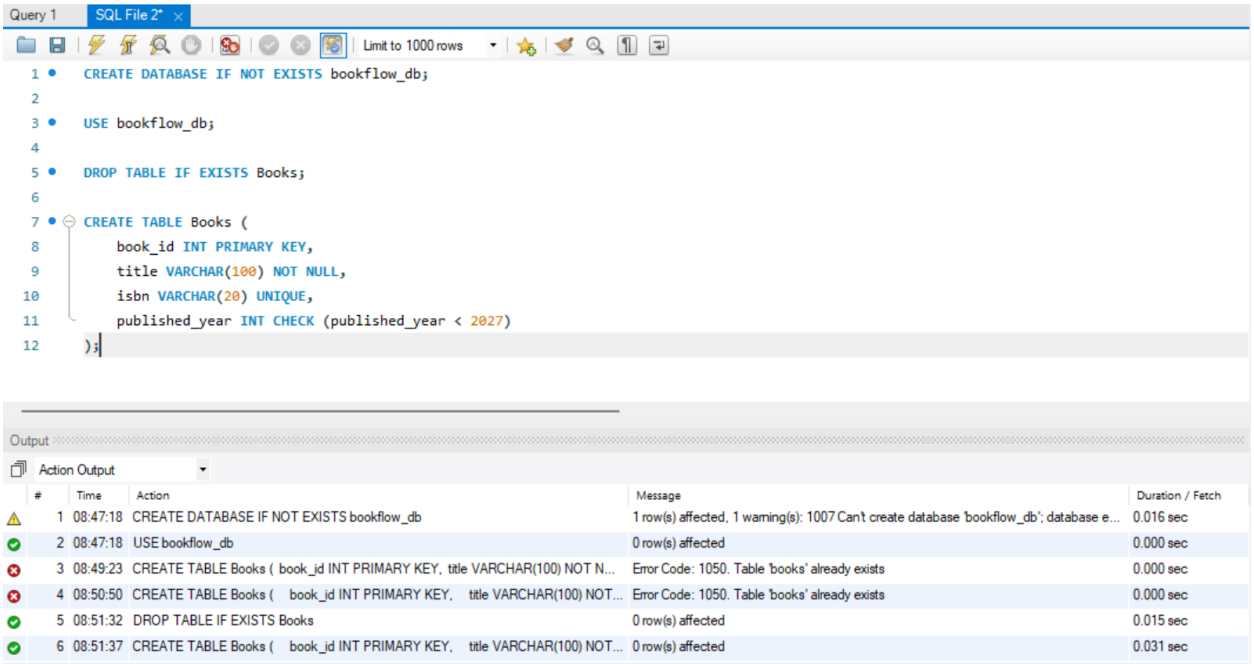
Week-2

Name : S. Manvi Vaishnavi
Roll Number : 2520030326

Step 1 — Create Database



Step 2 Table Creation (DDL) with Constraints :



Step 3 & 4 — Insert Books and Display

```
14 •  INSERT INTO Books (book_id, title, isbn, published_year) VALUES
15    (1, 'The Great Gatsby', '9780743273565', 1925),
16    (2, 'To Kill a Mockingbird', '9780061120084', 1960),
17    (3, '1984', '9780451524935', 1949);
18 •  SELECT * FROM Books;
19
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Books 1 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	08:49:23	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT ...	Error Code: 1050. Table 'books' already exists	0.000 sec
4	08:50:50	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) N...	Error Code: 1050. Table 'books' already exists	0.000 sec
5	08:51:32	DROP TABLE IF EXISTS Books	0 row(s) affected	0.015 sec
6	08:51:37	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) N...	0 row(s) affected	0.031 sec
7	08:52:58	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	08:52:58	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 5, 6 & 7 — Members Table: Create, Insert, Display

Query 1 | SQL File 2* | Limit to 1000 rows

```
28 •  INSERT INTO Members (member_id, full_name, email) VALUES
29    (101, 'Rohini', 'rohini.smith@email.com'),
30    (102, 'Rahasyaa', 'rahasyaa.wilson@email.com'),
31    (103, 'Pranitha', 'pranitha.brown@email.com');
32
33 •  SELECT * FROM Members;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

member_id	full_name	email
101	Rohini	rohini.smith@email.com
102	Rahasyaa	rahasyaa.wilson@email.com
103	Pranitha	pranitha.brown@email.com
NULL	NULL	NULL

Result Grid | Form Editor

Members 3 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	08:58:54	DELETE FROM Members WHERE member_id IN (1, 2, 3, 4, 5, 101, 102, 103)	8 row(s) affected	0.000 sec
14	08:58:59	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Rohini', 'rohi...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
15	08:58:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Step 8 — Create Loans Table (FOREIGN KEYS

```
34
35 • CREATE TABLE Loans (
36     loan_id INT PRIMARY KEY,
37     member_id INT,
38     book_id INT,
39     loan_date DATE,
40     FOREIGN KEY (member_id) REFERENCES Members(member_id),
41     FOREIGN KEY (book_id) REFERENCES Books(book_id)
42 );
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	08:58:59	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Rohini', 'rohi...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
15	08:58:59	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
16	09:00:36	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id I...	0 row(s) affected	0.031 sec

Step 9 & 10 — Insert 10 Loans and Display

Query 1 | SQL File 2*

Limit to 1000 rows

```
44 • INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
45     (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),
46     (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),
47     (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),
48     (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),
49     (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
50 • SELECT * FROM Loans;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

Result Grid | Form Editor | Field Types

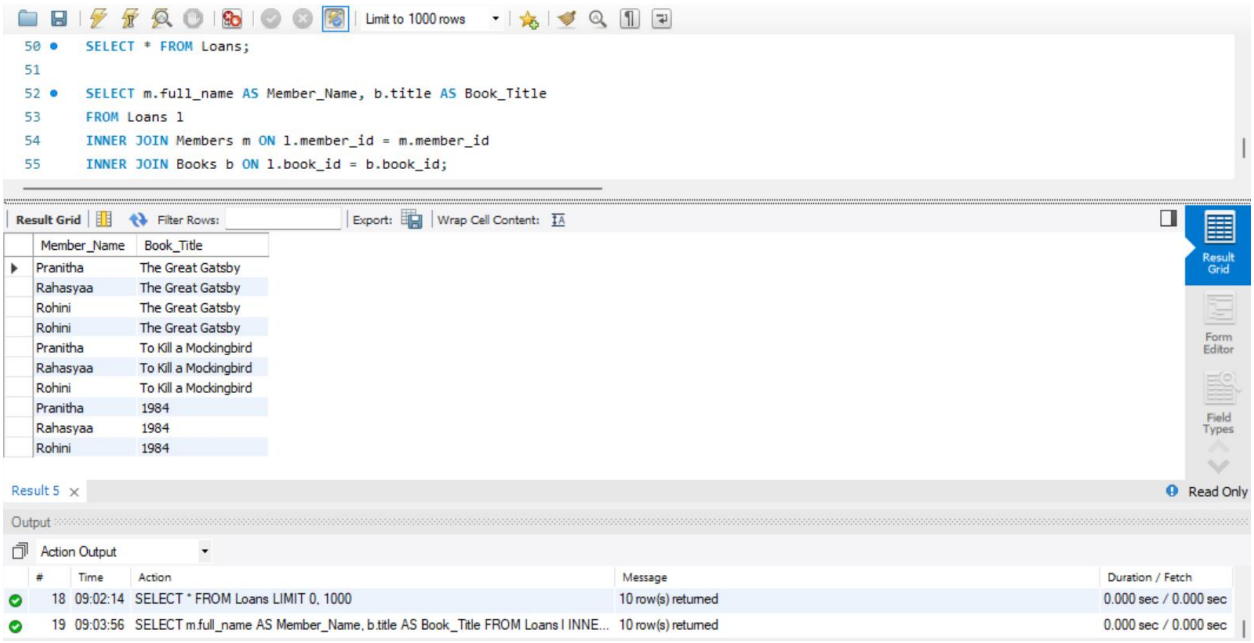
Loans 4 | Apply | Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
17	09:02:14	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1,...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
18	09:02:14	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Step 11 — JOIN Query (Who Borrowed What)



```
50 •  SELECT * FROM Loans;
51
52 •  SELECT m.full_name AS Member_Name, b.title AS Book_Title
53    FROM Loans l
54    INNER JOIN Members m ON l.member_id = m.member_id
55    INNER JOIN Books b ON l.book_id = b.book_id;
```

Result Grid

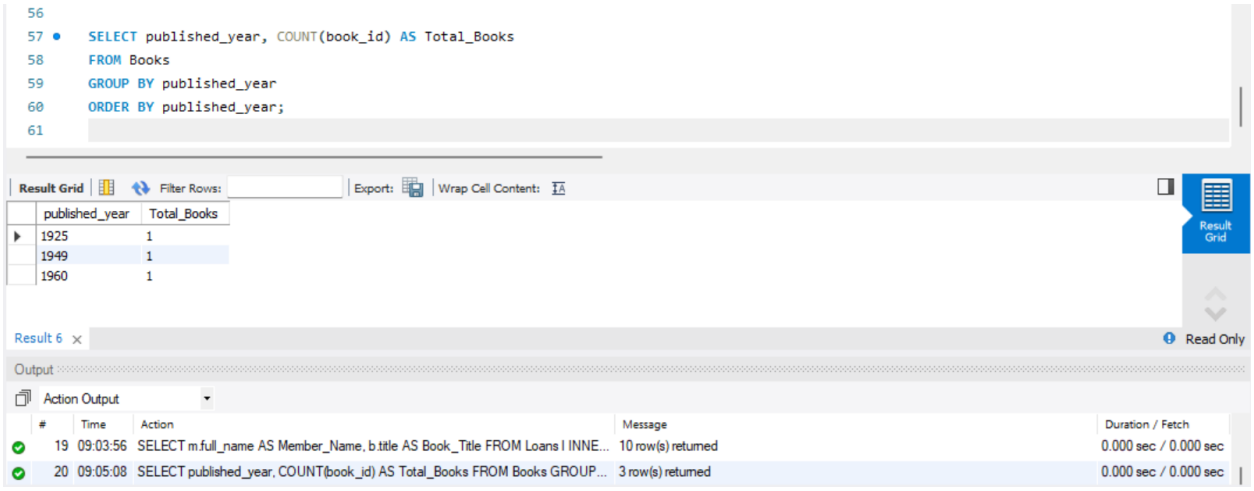
Member_Name	Book_Title
Pranitha	The Great Gatsby
Rahasyaa	The Great Gatsby
Rohini	The Great Gatsby
Rohini	The Great Gatsby
Pranitha	To Kill a Mockingbird
Rahasyaa	To Kill a Mockingbird
Rohini	To Kill a Mockingbird
Pranitha	1984
Rahasyaa	1984
Rohini	1984

Result 5 — Read Only

Output — Action Output

#	Time	Action	Message	Duration / Fetch
18	09:02:14	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
19	09:03:56	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNE...	10 row(s) returned	0.000 sec / 0.000 sec

Step 12 — GROUP BY Query (Books per Year)



```
56
57 •  SELECT published_year, COUNT(book_id) AS Total_Books
58    FROM Books
59    GROUP BY published_year
60    ORDER BY published_year;
61
```

Result Grid

published_year	Total_Books
1925	1
1949	1
1960	1

Result 6 — Read Only

Output — Action Output

#	Time	Action	Message	Duration / Fetch
19	09:03:56	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNE...	10 row(s) returned	0.000 sec / 0.000 sec
20	09:05:08	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP...	3 row(s) returned	0.000 sec / 0.000 sec

Step 13–16 — Donation via TRANSACTION (Atomicity)

```
61
62 • CREATE TABLE Donation_History (
63     donation_id INT PRIMARY KEY,
64     book_id INT,
65     donor_name VARCHAR(100),
66     donation_date DATE,
67     FOREIGN KEY (book_id) REFERENCES Books(book_id)
68   );
69 • START TRANSACTION;
70 • INSERT INTO Books (book_id, title, isbn, published_year)
71   VALUES (4, 'Animal Farm', '9780451526342', 1945);
72 • INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
73   VALUES (1, 4, 'Raj Kumar', CURDATE());
74 • COMMIT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	09:06:29	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected	0.000 sec
26	09:06:29	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

```
76 • CREATE INDEX idx_books_isbn ON Books(isbn);
77 • SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Result Grid

Books 7 ×

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	09:07:38	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
28	09:07:38	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec